

UNIVERSITY COLLEGE OF JAFFNA

UNIVERSITY OF VOCATIONAL TECHNOLOGY

HIGHER NATIONAL DIPLOMA IN INFORMATION AND COMMUNICATION TECHNOLOGY

NATIONAL VOCATIONAL QUALIFICATION LEVEL 06

Project planning documents
Online Food Ordering and Management System

Software Change Management (K72C001M12)

Assessment No 01

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PROJECT PLANNING DOCUMENT

Online Food Ordering and Management System

Project Name: Online Food Ordering and Management System
Document Name: Project Planning Document
Document Version: 1.0
Project Type: Web-Based System
Project Status: Planned

1. Project Introduction

The Online Food Ordering and Management System is a web-based system developed to make food ordering and restaurant order management easier.

Customers can view food items, search the menu, add food to a cart, place orders, provide delivery details, select a payment method, and check order status.

Restaurant staff can manage food items, prices, availability, and customer orders.

The administrator can manage users and system access.

This project plan explains how the project will be planned, developed, tested, and completed.

2. Project Purpose

The purpose of this project is to develop a web-based food ordering system that provides a simple way for customers to order food and for restaurant staff to manage food and orders.

The project will provide separate functions for:

- Customers
- Administrator

The system will help reduce mistakes that can happen when orders are handled through phone calls, social media messages, or direct communication.

3. Project Objectives

The main objectives of the project are:

1. Develop a web-based food ordering system.
2. Allow customers to register and login.
3. Allow customers to view available food.
4. Allow customers to search for food.
5. Allow customers to view food details and prices.
6. Allow customers to add food to a cart.
7. Allow customers to place food orders.
8. Allow customers to provide delivery details.
9. Allow customers to select a payment method.
10. Allow customers to check order status.
11. Allow restaurant staff to manage food items.
12. Allow restaurant staff to manage food prices.
13. Allow restaurant staff to manage food availability.
14. Allow restaurant staff to manage customer orders.
15. Allow administrators to manage users and access.
16. Test the system before final submission.
17. Complete the project within the planned time.

4. Project Scope

4.1 Included in the Project

The project will include:

Customer Module

- Customer registration
- Customer login
- Customer account
- Food browsing
- Food search
- Food categories
- Food details
- Shopping cart
- Quantity management
- Order placement
- Delivery details
- Payment method selection
- Order confirmation
- Order history

- Order status

Restaurant Staff Module

- Staff login
- Food management
- Add food
- Edit food
- Delete food
- Price management
- Food availability
- View customer orders
- Update order status

Administrator Module

- Administrator login
- User management
- User role management
- User access management
- View orders

5.2 Not Included in the Project

The following functions will not be developed:

- GPS tracking
- Live delivery tracking
- Real-time customer location tracking

Customers will enter their delivery information manually.

6. Project Method

The project will be developed using an **incremental development approach**.

The system will be divided into smaller parts and developed step by step.

The main stages are:

1. Project planning
2. Requirement gathering
3. System analysis
4. System design

5. Database design
6. User interface design
7. Development
8. Module testing
9. System integration
10. System testing
11. Bug fixing
12. Documentation
13. Final review
14. Final submission

This approach allows problems to be found earlier instead of waiting until the whole system is completed.

7. Project Development Phases

Phase 1: Project Planning

Activities

- Define the project
- Identify the problem
- Define project objectives
- Define project scope
- Identify users
- Prepare project plan
- Identify risks

Output

- Project Planning Document
- Initial project scope
- Initial schedule

Phase 2: Requirements Analysis

Activities

- Identify customer requirements
- Identify restaurant staff requirements
- Identify administrator requirements
- Define functional requirements
- Define non-functional requirements

- Define business rules
- Define user stories
- Define acceptance criteria

Output

- Requirements Documentation
- Requirements Baseline

Phase 3: System Analysis and Design

Activities

- Analyse system requirements
- Identify system modules
- Design system workflow
- Design database structure
- Identify database tables
- Design relationships between tables
- Design system interfaces
- Design user roles and access

Output

- System design
- Database design
- System workflow
- Interface designs

Phase 4: Database Development

Activities

- Create the database
- Create required tables
- Define primary keys
- Define foreign keys
- Define relationships
- Add required data
- Test database operations

Main Data Areas

- Users

- Customers
- Food Items
- Categories
- Orders
- Order Items
- Payments

Output

- Working database
- Database tables
- Database relationships

Phase 5: Customer Module Development

Activities

Develop the customer functions:

1. Registration
2. Login
3. View food
4. Search food
5. View food details
6. Add food to cart
7. Update cart
8. Remove cart items
9. Enter delivery details
10. Select payment method
11. Place order
12. View order history
13. Check order status

Output

- Working Customer Module

Phase 6: Restaurant Staff Module Development

Activities

Develop the restaurant staff functions:

1. Staff login
2. Food management
3. Add food
4. Edit food
5. Delete food
6. Update price
7. Update availability
8. View customer orders
9. Update order status

Output

- Working Restaurant Staff Module

Phase 7: Administrator Module Development

Activities

Develop the administrator functions:

1. Administrator login
2. View users
3. Manage users
4. Manage user roles
5. Manage user access
6. View orders

Output

- Working Administrator Module

Phase 8: Module Testing

Each module will be tested separately.

Customer Module Testing

Test:

- Registration
- Login
- Food search
- Food display
- Cart

- Order placement
- Delivery information
- Payment selection
- Order history
- Order status

Restaurant Module Testing

Test:

- Food adding
- Food editing
- Food deleting
- Price updating
- Availability updating
- Order viewing
- Order status updating

Administrator Module Testing

Test:

- Login
- User management
- Role management
- Access control
- Order viewing

Phase 9: Integration Testing

After individual modules are tested, the modules will be connected and tested together.

Examples:

Customer → Restaurant

Customer places an order.

↓

Restaurant staff receives the order.

Restaurant → Customer

Restaurant staff updates the order status.

↓

Customer can see the updated status.

Customer → Database

Customer creates an account.

↓

Customer information is saved in the database.

Restaurant → Database

Restaurant staff updates food information.

↓

Updated food information is saved in the database.

10. System Testing

The complete system will be tested after the modules are connected.

Testing will check:

- Main functions
- User login
- User roles
- Food management
- Cart
- Order process
- Payment selection
- Order status
- Data validation
- Database operations
- Security
- Error messages
- System navigation

11. Project Work Breakdown Structure

1. Project Management

1.1 Project Planning

1.2 Requirement Planning

1.3 Time Planning

1.4 Risk Planning

1.5 Progress Monitoring

2. Requirements

2.1 Identify Stakeholders

2.2 Identify User Roles

2.3 Gather Requirements

2.4 Define Functional Requirements

2.5 Define Non-Functional Requirements

2.6 Prepare Requirements Documentation

3. System Design

3.1 System Architecture

3.2 System Workflow

3.3 Database Design

3.4 Interface Design

3.5 User Role Design

4. Development

4.1 Customer Module

4.2 Restaurant Staff Module

4.3 Administrator Module

4.4 Database Integration

4.5 Validation

4.6 Access Control

5. Testing

5.1 Unit Testing

5.2 Integration Testing

5.3 System Testing

5.4 Validation Testing

5.5 Bug Fixing

5.6 Retesting

6. Documentation

6.1 Requirements Documentation

6.2 Design Documentation

6.3 Testing Documentation

6.4 User Documentation

6.5 Final Project Report

7. Final Submission

7.1 Final Review

7.2 Final Testing

7.3 Project Demonstration

7.4 Final Submission

12. Project Schedule

The following is the planned schedule for the project.

Phase	Main Activities	Planned Duration
1	Project Planning	1 Week
2	Requirements Analysis	1 Week
3	System Analysis and Design	2 Weeks
4	Database Development	1 Week
5	Customer Module	2 Weeks
6	Restaurant Staff Module	2 Weeks
7	Administrator Module	1 Week
8	Integration	1 Week
9	Testing	2 Weeks
10	Bug Fixing and Retesting	1 Week
11	Documentation	1 Week
12	Final Review and Submission	1 Week

Total Planned Duration: 16 Weeks

13. Project Milestones

Milestone	Expected Result
M1	Project planning completed
M2	Requirements completed
M3	System design completed
M4	Database completed
M5	Customer module completed
M6	Restaurant staff module completed
M7	Administrator module completed

M8	Modules integrated
M9	System testing completed
M10	Major bugs fixed
M11	Documentation completed
M12	Final project completed

14. Project Resources

14.1 Human Resources

The project requires:

- Student / Developer
- Project Supervisor
- Tester

For an individual academic project, the student may perform most development and testing activities.

The supervisor will provide guidance and review the project progress.

14.2 Software Resources

The project may require:

- Visual Studio / Visual Studio Code
- Git
- GitHub
- Database management software
- Web browser
- Testing tools
- Diagram/design tools

The exact software used will depend on the final development setup.

14.3 Hardware Resources

The project requires:

- Computer or laptop
- Internet connection
- Keyboard

- Mouse
- Sufficient storage

15. Project Roles and Responsibilities

Role	Responsibilities
Student / Developer	Planning, analysis, design, coding, testing, documentation
Supervisor	Guidance, review, feedback
Tester	Test system functions and report problems
Restaurant Staff User	Provide requirements related to food and order management
Customer User	Provide requirements related to food ordering

16. Risk Management

Risks are problems that may affect the project.

Risk	Possible Effect	Action
Requirement changes	Development delay	Review requirements before development
Lack of time	Project may not be completed	Follow the project schedule
Coding errors	Functions may not work	Test each module
Database errors	Data may not be saved correctly	Test database operations
Integration errors	Modules may not work together	Perform integration testing
Data loss	Project work may be lost	Use Git and GitHub regularly
Internet problems	Development or online services may be affected	Keep local copies of project files
Payment integration problems	Online payment may not work	Test payment functions early
Security problems	User data may be at risk	Apply login and access control
Scope changes	More work than planned	Follow the approved requirements
Testing delays	Bugs may remain	Start testing during development

17. Risk Response Plan

Requirement Changes

If requirements change, the change will be reviewed before development.

The effect on time, database, development, and testing will be checked.

Data Loss

The project will be stored using Git and GitHub.

Regular commits will be made during development.

Development Problems

Development problems will be recorded and fixed before moving to the next major stage.

Testing Problems

Failed tests will be recorded.

The related function will be corrected and tested again.

18. Quality Plan

The project quality will be checked during development.

The following activities will be used:

1. Review requirements before development.
2. Use clear naming for database tables and system components.
3. Validate user input.
4. Test each module separately.
5. Test modules together.
6. Test the complete system.
7. Record errors.
8. Fix identified errors.
9. Test the fixed functions again.
10. Review the final system before submission.

19. Testing Plan

Unit Testing

Individual functions will be tested.

Examples:

- Login
- Registration
- Food search
- Add to cart
- Add food
- Update food
- Order status update

Integration Testing

Connected modules will be tested.

Examples:

- Customer order → Restaurant order
- Restaurant status → Customer status
- Customer data → Database
- Food data → Customer menu

System Testing

The complete system will be tested from beginning to end.

Example:

Customer Login

↓

Search Food

↓

Add Food to Cart

↓

Place Order

↓

Restaurant Receives Order

↓

Restaurant Updates Status

↓

Customer Checks Status

20. Change Management

Project changes will be controlled.

If a new feature is requested:

1. Record the requested change.
2. Check whether it is inside the project scope.
3. Check the effect on project time.
4. Check the effect on the database.
5. Check the effect on development.
6. Check the effect on testing.
7. Approve or reject the change.
8. Update the project documents if the change is approved.

Changes that are not necessary for the main system should not be added if they can delay the project.

21. Communication Plan

Project communication will be used to track progress and problems.

Communication	Purpose	Frequency
Supervisor Meeting	Discuss project progress	Weekly
Progress Update	Report completed work	Weekly

Requirement Review	Check requirements	During analysis
Testing Review	Discuss testing problems	During testing
Final Review	Check completed system	Before submission

22. Project Progress Tracking

Project progress will be checked using the following status:

Status	Meaning
Not Started	Work has not started
In Progress	Work is currently being done
Completed	Work has been completed
Testing	Work is being tested
Rework	Problems were found and changes are needed
Approved	Work has been checked and accepted

23. Project Deliverables

The main project deliverables are:

1. Project Planning Document
2. Requirements Documentation
3. System Design
4. Database Design
5. User Interface Design
6. Source Code
7. Database
8. Test Plan
9. Test Cases
10. Test Results
11. User Guide
12. Final Project Report
13. GitHub Repository
14. Final Working System

24. Document Management

Project documents will be stored in an organized project folder.

Example:

```
Online-Food-Ordering-Management-System
│
├── Requirements
│   └── Requirements-Documentation
│
├── Planning
│   └── Project-Planning
│
├── Design
│   ├── System-Design
│   ├── Database-Design
│   └── UI-Design
│
├── Development
│   └── Source-Code
│
├── Database
│   └── Database-Files
│
├── Testing
│   ├── Test-Plan
│   ├── Test-Cases
│   └── Test-Results
│
└── Documentation
    └── Final-Report
```

25. Version Control Plan

Git and GitHub will be used to manage project source code and version history.

The project will use regular commits.

Example commit messages:

- Initial project setup
- Added customer registration
- Added customer login
- Added food management
- Added shopping cart
- Added order management
- Added administrator module
- Added database connection

- `Added validation`
- `Fixed order status issue`
- `Added testing files`

The GitHub repository will be used as the main online location for the project source code and related project files.

26. Project Monitoring

Project progress will be checked against the planned schedule.

The following items will be monitored:

- Completed tasks
- Remaining tasks
- Development progress
- Testing progress
- Requirements changes
- Project risks
- Problems found
- Time used
- Final submission date

If a task is delayed, the reason will be recorded and the remaining schedule will be reviewed.

27. Completion Criteria

The project will be considered complete when:

- Customer registration works.
- Customer login works.
- Food items can be viewed.
- Food search works.
- Food details can be viewed.
- Food can be added to the cart.
- Cart quantities can be changed.
- Cart items can be removed.
- Customers can place orders.
- Delivery details can be entered.
- Payment method can be selected.
- Customers can view order status.
- Restaurant staff can manage food.
- Restaurant staff can manage food availability.

- Restaurant staff can manage orders.
- Restaurant staff can update order status.
- Administrator can manage users.
- User roles work correctly.
- Input validation works.
- Main database operations work.
- Integration testing is completed.
- System testing is completed.
- Major errors are fixed.
- Required documentation is completed.
- Final project is reviewed before submission.

28. Project Constraints

The project has the following constraints:

1. The project must be completed within the planned academic period.
2. Development time is limited.
3. Available resources may be limited.
4. The system depends on a working database.
5. Online payment may depend on an external payment service.
6. GPS and live delivery tracking are outside the project scope.
7. Major changes to requirements may increase development time.
8. The project must be tested before final submission.

29. Project Assumptions

The project is based on these assumptions:

1. The required development tools are available.
2. The developer has access to a computer.
3. Internet access is available when required.
4. The database can be installed and used during development.
5. Users provide correct information.
6. Restaurant staff update food and order information correctly.
7. The project requirements remain mostly stable after the baseline.
8. The required project time is available for development and testing.

30. Project Communication and Reporting

Project progress will be recorded regularly.

Each progress update should contain:

- Work completed
- Work currently in progress
- Problems found
- Problems solved
- Work planned next
- Any requirement changes
- Any schedule changes

This will help to identify problems early and keep the project on schedule.

31. Final Project Review

Before final submission, the following areas will be reviewed:

Requirements

- Check whether all required functions are included.

Development

- Check whether the main modules work correctly.

Database

- Check whether data is saved and retrieved correctly.

Testing

- Check whether the required tests have been completed.

Security

- Check login and user access.

Documentation

- Check whether project documents are complete.

GitHub

- Check source code and project files.
- Check README.
- Check. gitignore.
- Check commit history.

- Check that unnecessary files are not uploaded.

32. Final Deliverables Checklist

- [ ] Working Online Food Ordering and Management System
- [ ] Customer Module
- [ ] Restaurant Staff Module
- [ ] Administrator Module
- [ ] Database
- [ ] Requirements Documentation
- [ ] Project Planning Document
- [ ] System Design
- [ ] Database Design
- [ ] Test Plan
- [ ] Test Cases
- [ ] Test Results
- [ ] User Guide
- [ ] Final Report
- [ ] GitHub Repository
- [ ] README
- [ ] Final project review

33. Project Completion

The project will be completed after the planned system functions are developed, tested, and reviewed.

The final system must allow customers to order food and check order status.

Restaurant staff must be able to manage food and customer orders.

Administrators must be able to manage users and access.

All major errors found during testing must be fixed before final submission.

The final source code, database files, testing documents, and other required project documents will be maintained in the project repository and project folders.

34. Conclusion

This Project Planning Document provides the plan for developing the Online Food Ordering and Management System.

The project will start with planning and requirements analysis and then continue through system design, database development, module development, testing, documentation, and final submission.

The project will be developed in smaller stages so that each part can be checked before moving to the next stage.

The main focus of the project is to provide a working food ordering system for customers and a food and order management system for restaurant staff, with administrator functions for user and access management.

The project will follow the approved requirements and planned schedule. Changes will be recorded and reviewed before they are added to the project.